

dFrom Spikes to Symptoms: Simulating SC-Driven Multisensory Deficits in Autism Spectrum Disorder

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ABSTRACT

Individuals with autism spectrum disorder (ASD) frequently exhibit altered multisensory integration, particularly enlarged temporal binding windows that impair speech perception and social communication. While various synaptic and cellular alterations have been implicated in ASD, the causal pathways linking these circuit-level perturbations to specific behavioural phenotypes remain elusive. To address this gap, we developed a biologically plausible spiking neural network of the superior colliculus, a key subcortical hub for audiovisual binding, incorporating realistic alpha-amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid and N-methyl-D-aspartate (NMDA) receptor kinetics, γ -aminobutyric acid-mediated inhibition, and spike-timing-dependent plasticity. Through systematic parameter manipulation, we show that distinct neural alterations produce characteristic integration deficits: reduced neuronal adaptation and diminished feedforward inhibition uniformly expanded both spatial and temporal binding windows to varying degrees, indicating distinct roles for early inhibitory gating in space and time. Furthermore, NMDA conductance alterations bidirectionally modulated these windows, with hyperfunction generating permissive fusion and hypofunction curtailing integration. Each perturbation uniquely affected localization precision and multisensory benefit. Together, these results show how heterogeneous synaptic and cellular changes can converge on similar multisensory symptoms while leaving unique computational fingerprints. The framework yields testable predictions - linking, for example, restoration of adaptation or targeted augmentation of inhibitory control to normalization of binding - and suggests mechanism-specific avenues for intervention. By mapping synaptic alterations to measurable perceptual outcomes in a biologically grounded circuit, our work clarifies potential sources of phenotypic variability in autism and outlines principled targets for future experimental and translational work.

INTRODUCTION

Our experience of the world is constructed from sensory inputs that are inherently partial and frequently ambiguous. To form coherent perceptions, the brain integrates multiple signals, each governed by distinct constraints. For example, determining the shape of a nearby object may involve combining various visual cues — such as perspective, shading, and texture — as well as inputs from other senses, like size information from both vision and touch (Ernst & Banks, 2002; Georgieva et al., 2008; Norman et al., 2003). This integration helps to resolve ambiguities and can lead to faster and more precise judgments (Hagmann & Russo, 2016).

The ubiquitous nature of multisensory integration in neural processing renders it particularly susceptible to the disruptive effects of neurodevelopmental disorders. One of the most prevalent and complex of these disorders, which is associated with multiple multisensory integration deficits, is autism spectrum disorder (ASD; Feldman et al., 2018). Population studies indicate that ~90% of autistic individuals report hyper- or hypo-reactivity across sensory modalities (Balasco et al., 2020). In particular, psychophysical work has shown that individuals with ASD have enlarged spatiotemporal binding windows (i.e., broadened temporal binding windows and reduced precision in spatial localization) for audiovisual stimuli (de Boer-Schellekens, Eussen, et al., 2013; Foss-Feig et al., 2010). Although widened multisensory binding windows are not unique to ASD and have also been reported in other conditions, they are robustly implicated in ASD and therefore provide a useful mechanistic phenotype for testing ASD-motivated circuit hypotheses in the superior colliculus (Wallace & Stevenson, 2014; Zhou et al., 2018). These sensory irregularities could potentially accumulate across modalities, leading to effects which could compound and affect higher level processing which could in turn impact daily functioning, making social environments overwhelming and leading to behaviours such as withdrawal or distress (Kawakami & Otsuka, 2021; Thye et al., 2018).

The superior colliculus (SC) is a brain area situated in the midbrain tectum, and has emerged as a key region for understanding multisensory integration deficits in ASD (Jure, 2019). The SC functions as a principal node for audiovisual integration, receiving convergent inputs from multiple sensory modalities (Benavidez et al., 2021). Within the SC, multisensory neurons receive congruent retinal and brainstem auditory afferents, aligning cross-modal inputs originating from common external events

(Meredith & Stein, 1986). This part of the brain permits robust cross-modal enhancements of responses detectable by both human neuroimaging studies and animal electrophysiology (Calvert et al., 2001; Stein & Rowland, 2020). Notably, the neural responses of this part of the brain are maximal when audiovisual events co-occur during a narrow time window, providing a cellular correlate for the coincidence detection underlying multisensory binding (Meredith et al., 1987). The computational processes key to such temporal summation depends on a delicate balance between fast α -amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid (AMPA) receptor-mediated excitation, slower N-methyl-D-aspartate (NMDA) receptor-mediated excitation, and γ -aminobutyric acid (GABA)ergic inhibition (Binns & Salt, 1994). Increasing developmental and clinical evidence supports the view that these integrative gains for such binding are attenuated in ASD, providing a robust case for a relationship between early dysfunction of the SC circuitry and the typical phenotypes of the senses evident in autism (Siemann et al., 2020).

Deficits of multisensory integration characteristic of ASD correspond with disruptions of the SC, whereby audiovisual responses are decreased both in autistic groups and mouse models of the disorder (Jure, 2019; Siemann et al., 2017). Postmortem, genomic, and pharmacological data demonstrate that altered NMDA expression, particularly GluN2B reductions, are associated with atypical audiovisual temporal and spatial binding windows (Bahry et al., 2021; Lee et al., 2024; Wallace & Stevenson, 2014). Electrophysiological research also uncovered decreased sensory adaptation and elevated excitation-inhibition (E/I) ratio of SC circuits, with each of these perturbations accompanying corresponding internal shifts in sensory processing capabilities (Rubenstein & Merzenich, 2003; Turi et al., 2016). Despite substantial associative evidence, direct causal pathways between specific circuit perturbations and sensory response alterations have not yet been established.

Traditional experimental methods face inherent limitations in establishing these causal mechanistic links (Wolff & Ölveczky, 2018). Single unit electrophysiology, psychophysics, and focal lesion studies have identified associations between neural activity and multisensory behaviour (Meredith & Stein, 1983; Stein & Stanford, 2008). But these approaches are not typically capable of establishing causality, as they sample only fragments of distributed circuits. In addition to this, it is experimentally challenging to vary synaptic parameters in living tissues (Reid et al., 2019; Wolff & Ölveczky, 2018). As a result, using these methods to link alterations in spatiotemporal

binding window metrics in ASD to specific collicular mechanisms remains largely infeasible (Crosse et al., 2022). Computational models seek to overcome this limitation through controlled parameter sweeps that enable identification of causal links between circuit perturbations and behaviour (Jun et al., 2021), producing hypotheses of SC function that can be tested experimentally.

Current computational models of multisensory integration have provided valuable theoretical advances while also exposing critical methodological limitations restricting their application for explaining ASD-related dysfunction on a level of neural circuits (Loosen et al., 2025; Stein et al., 2014). These approaches act at various levels of analysis, each of which yields unique advances but also significant gaps in understanding. At a statistical level, Bayesian causal inference models demonstrate how the brain optimally balances sensory evidence in order to decide whether signals arise from common or independent sources (Lerousseau et al., 2022); a computation that appears to be altered in ASD (Sapey-Triomphe et al., 2023). At a biophysically informed generative-modeling level, dynamic causal modelling uses fMRI data to estimate directed connectivity between brain regions during multisensory tasks, revealing how effective connectivity changes with stimulus conditions (Bouton et al., 2020). Meanwhile at the neural implementation level, neural networks such as MultiNet demonstrate how this causal inference arises from populations of neurons which have congruent and opposite tuning features, akin to those which exist for macaque MSTd cortex (Rideaux et al., 2021). While these models provide useful insights into multisensory integration, they typically omit explicit representations of receptor kinetics and other synapse specific dynamics (Barri et al., 2022; Stein et al., 2014), which limits their capacity to trace neurotransmitter level perturbations directly to behavioural observations (Barri et al., 2022; Gjorgjieva et al., 2016). Defining these mechanistic links is key to explaining how synaptic abnormalities in the SC cause the multisensory symptoms seen in ASD.

Biologically plausible spiking neural networks (SNNs) represent a promising approach to address these shortcomings by incorporating the detailed synaptic mechanisms that are missing from current models (Agnes & Vogels, 2024; Saponati & Vinck, 2023). SNNs can embed explicit AMPA and NMDA conductance dynamics, GABAergic inhibition, and spike-timing-dependent plasticity (STDP) at millisecond temporal resolution (Brunel & Wang, 2003; Maass et al., 2002; Wang, 1999). Detailed modelling of these biological features allows synapse-specific hypotheses about ASD-

related dysfunction to be tested computationally. Rate-coded SC models, such as those developed by Cuppini et al. (2011), have successfully reproduced key features of multisensory integration including superadditive audiovisual gain, demonstrating the potential explanatory power of network-based approaches with detailed spiking-level implementations. Large-scale integrate-and-fire reconstructions of SC circuitry have also confirmed that SNN architectures can accurately model the fundamentals of collicular circuits (Alizadeh & Van Opstal, 2022). However, SNNs modelling multisensory integration processes in the SC have not yet been used to interrogate ASD-related circuit anomalies. This represents a clear opportunity to extend biologically grounded computational frameworks to test how NMDA receptor dysregulation, altered E/I balance, or other synaptic perturbations could reshape multisensory tuning properties in the SC and explain perceptual impairments associated with ASD.

Here we introduce a biologically constrained SNN model of the SC that allows NMDA conductance, E/I balance, and synaptic adaption parameters to be independently manipulated and systematically varied. We therefore interpret the phenotypes modelled here as ASD-relevant mechanistic phenotypes rather than ASD-exclusive diagnostic signatures. Through focused parameter sweeps, we show that specific changes in synaptic properties lead to evidence of multisensory integration difficulties similar to those seen in persons with ASD. By isolating these causal pathways under tightly controlled conditions, the model makes testable predictions regarding how unique circuit alterations may accumulate to generate the sensory processing differences present in ASD. By manipulating these causal mechanisms under tightly controlled conditions, the model makes predictions linking specific network perturbations to the disruptions in behaviour seen in autism.

METHODS

Network architecture. We constructed a two-layer spiking neural network comprising auditory and visual input layers (180 excitatory neurons each, spanning azimuth 0° – 179°) that projected to multisensory excitatory (180 neurons) and inhibitory (54 neurons) layers. The inhibitory fraction (30%) was chosen to approximate typical interneuron densities. (Essig et al., 2021; Feldmeyer, 2012).

To model the individual neurons' spiking behaviour, we used the Izhikevich neuron, which couples a quadratic membrane equation with a recovery variable and an after-spike reset to reproduce the spiking patterns observed in SC recordings, while maintaining ease of computation for large-scale simulations (Izhikevich, 2003). The state equations of the neuron model are

$$\begin{aligned}\frac{dv}{dt} &= 0.04v^2 + 5v + 140 - u + I(t), \\ \frac{du}{dt} &= a(bv - u),\end{aligned}$$

with the spike/reset condition

$$v \geq 30\text{mV} \Rightarrow \{ v \leftarrow c, u \leftarrow u + d \}$$

where v is membrane potential, u is a recovery variable and $I(t)$ denotes the net synaptic input current at time t . In this formulation, a sets the time scale of recovery (smaller $a \rightarrow$ slower adaptation), b controls how strongly u couples to v (restorative feedback near rest), c is the post-spike reset potential, and d is the after-spike increment to the recovery variable. An added advantage of the Izhikevich model was that we could switch between several neuron behavioural modes by controlling a few key parameters. For example, regular-spiking dynamics ($a=0.02$, $b=0.2$, $c=-65$, $d=8$) captured the sustained firing patterns of excitatory projection neurons, while fast-spiking parameters ($a=0.1$, $b=0.2$, $c=-65$, $d=2$) reproduced the rapid, non-adapting responses characteristic of inhibitory interneurons that mediate feedforward inhibition in the SC. This distinction proved to be important for establishing the timing relationships between excitation and inhibition that underlie multisensory integration windows (Gingras et al., 2009).

The network operated at millisecond resolution ($dt = 0.1$ ms) and all simulations were implemented in native PyTorch, with 100 sub-steps per frame (each frame was 10 ms long), as this level of temporal detail enables fine-scale coincidence detection that distinguishes true multisensory events from false correlations (Wu et al., 2025;

Zenke & Gerstner, 2014). This high temporal resolution also allowed synaptic current computations to evolve realistically and enabled STDP mechanisms to shape connectivity based on the correct temporal relationships between pre- and postsynaptic activity (Shouval et al., 2010).

Training protocol and stimulus generation. Networks were trained unsupervised for 80 epochs (1,000 sequences per epoch; 20 frames/200 ms per sequence). Each sequence contained up to two 50-ms events with probabilities 60% congruent audiovisual, 20% auditory-only, and 20% visual-only; during bimodal events, each modality was independently jittered by $\pm 3^\circ$ with 60% probability. Inputs were converted to Poisson spike trains (0–300 Hz, 10-ms bins), with intensities ranging from 0.4 to 1.0 and additive Gaussian spatial noise ($\sigma = 0.02$) (De Gelder & Bertelson, 2003; Stevenson, Ghose, et al., 2014). A maximum of two events could occur simultaneously.

To ensure the network could tolerate minor deviations in spatial alignment, during each bimodal event the auditory and visual components were independently jittered with 60% probability by a small lateral offset drawn uniformly from $\pm 3^\circ$, such that either modality (or both) could be displaced while the other remained aligned. Before entering the unisensory A and V layers, each event profile was converted into a population of Poisson-distributed spikes (0–300 Hz, 10 ms bins), providing physiologically realistic spiking input for synaptic plasticity (Thieu & Melnik, 2025). Input intensities varied logarithmically between 0.4 and 1.0 to encourage contrast-invariant responses, and Gaussian spatial profiles were corrupted with additive zero-mean Gaussian noise ($\sigma = 0.02$, applied independently to every neuron and frame in both modalities) to simulate sensory uncertainty and avoid overfitting to noise-free inputs (Faisal et al., 2008; Hasinoff, 2014).

The network's operating regime required careful calibration to balance sensitivity against saturation. By trial and error, we identified an input scaling factor that elicited approximately ten multisensory spikes in response to a single unimodal pulse over a second, a level that permitted robust responses without ceiling effects. Feedforward inhibitory gain was set to match a physiologically accurate excitation/inhibition balance that supported both reliable detection and dynamic range for multisensory enhancement (Bhatia et al., 2019). Unless otherwise stated, behavioral assays were run in a near-silent baseline regime with no tonic background

drive beyond sensory noise. This yielded negligible pre-stimulus MSI firing. To validate ongoing variability and noise correlations, we additionally tested an independent Poisson background drive; the qualitative validation trends were preserved under this noisier regime.

Synaptic dynamics and receptor dynamics. Multisensory integration in the SC depends on the interactions between fast and slow excitatory currents that create temporal windows for coincidence detection (Binns & Salt, 1996; Meredith et al., 1987). To simulate such time-varying dynamics, we modelled dual-component excitatory synapses into our network, with physiologically constrained AMPA and NMDA conductances. AMPA receptors are responsible for fast synaptic transmission with a 2.5 ms decay constant, which gives them the precision needed to detect simultaneous events. In contrast, NMDA receptors exhibit slower kinetics with decay constants set to 80 ms, which leads to extended temporal activity, over which events farther apart in time could summate.

The NMDA conductance incorporated a voltage-dependent Mg^{2+} block, a key nonlinearity that makes these receptors function as coincidence detectors at the molecular level:

$$I_{NMDA} = g_{NMDA} m_{NMDA} \left(\frac{E_{NMDA} - V}{1 + e^{-0.062 \cdot (V + 35)}} \right)$$

where g_{NMDA} is the maximal synaptic conductance, m_{NMDA} is the fraction of receptors with ligand-gated channels open, V is the postsynaptic membrane potential, $1 + e^{-0.062 \cdot (V + 35)}$ is the voltage-dependent Mg^{2+} -block relief factor, and E_{NMDA} (≈ 0 mV) is the reversal potential driving the current. This formulation meant that NMDA receptors contributed minimal current at resting potentials but as neurons depolarized, these receptors became more active, which then created a positive feedback mechanism that amplified responses to coincident inputs (Trpevski et al., 2023; Zoicas & Kornhuber, 2019). The half-activation voltage of -35 mV and steepness factor of 0.062 mV⁻¹ matched measurements recorded from SC neurons.

The AMPA synapses were also modelled to display synaptic depression, following delayed relative firing of the pre- neurons. Short-term synaptic depression on AMPA-mediated connections followed the Tsodyks-Markram formula:

$$\frac{dR}{dt} = \frac{1 - R}{\tau_{rec}} - U R \delta_{(t - t_{spike})}$$

where R represents the fraction of available synaptic resources, $\tau_{rec} = 400$ ms is the recovery time constant, $U = 0.7$ is the utilization parameter determining the fraction of resources consumed per spike, and $\delta_{(t-t_{spike})}$ represents spike times. This activity-dependent attenuation prevents response saturation during sustained stimulation, as is observed in SC neurons (Lee et al., 2020). This short-term depression produced synaptic fatigue during repeated input and was distinct from the intrinsic spike-frequency adaptation controlled by the Izhikevich recovery variable. This depression manifested as the sensory adaptation displayed by neurons in the multisensory layer.

We included three inhibitory motifs: feedforward inhibition from unisensory to multisensory units, recurrent inhibition from the multisensory inhibitory population, and Mexican-hat lateral inhibition within the multisensory excitatory layer (local excitation radius = 4 neurons with surround suppression) (Cuppini et al., 2011). We designed this multi-scale inhibitory architecture to reproduce the centre-surround receptive fields characteristic of SC multisensory neurons.

Connectivity and topographic organisation. The SC plays a key role in the alignment of auditory and visual spatial maps even though they come from different topographic origins, which represents a fundamental goal of multisensory integration (King et al., 1988). We attempted to model this ability through structured connectivity patterns that promoted map alignment while allowing for flexibility during learning (Cuppini et al., 2011). The initial feedforward connections from unisensory to multisensory layers followed Gaussian spatial profiles centred on corresponding topographic positions (Ottes et al., 1986), providing a coarse scaffold for integration:

$$W_{ij} = A e^{\left[-\frac{(i-j)^2}{2\sigma^2}\right]}$$

where W_{ij} represents the connection strength from neuron j in the unisensory layer to neuron i in the multisensory layer, A is the peak amplitude, and $\sigma = 5$ neurons determine the neural radius of connectivity. These feedforward connections were present for both AMPA and NMDA synapses and remained plastic, allowing learning experience to shape the alignment between modalities.

To maintain Dale's principle while enabling gradient-based learning, we employed a soft-plus reparameterization for excitatory weights that ensured the network's weights would stay non-negative while maintaining the differentiability needed for learning (Wiemann et al., 2024).

$$W = \ln(1 + e^\theta) - \ln(2)$$

This formulation allowed for stable STDP-driven organization of topographic maps.

Neurophysiological studies demonstrate that auditory and visual information arrive at SC neurons with characteristic delays reflecting subcortical processing times (Keetels & Vroomen, 2012). We incorporated these timing differences through pathway-specific conduction delays: auditory signals were delayed by 10 ms and visual signals by 26 ms relative to multisensory neurons. These delays, derived from *in vivo* measurements of SC response latencies, created realistic temporal relationships between converging inputs and enabled the network to learn appropriate integration windows through STDP mechanisms (King, 2004).

Learning rules and homeostatic mechanisms. The development of aligned multisensory maps in the SC involves interactions between Hebbian plasticity, which strengthens correlated inputs, and homeostatic mechanisms that maintain stable firing rates and prevent pathological synchrony (Wu et al., 2020). We implemented this learning structure through multiple plasticity rules operating at distinct timescales (Daoudal & Debanne, 2003; Wu et al., 2020).

STDP learning shaped feedforward connections based on the temporal relationship between pre- and postsynaptic spikes (Vogels et al., 2013):

$$\Delta W = A_{stp} e^{-\frac{\Delta t}{\tau_{stp}}}, \quad \Delta t > 0$$

$$\Delta W = -A_{std} e^{\frac{\Delta t}{\tau_{std}}}, \quad \Delta t < 0$$

where ΔW is the weight change, Δt is the spike timing difference, $A_{stp} = A_{std} = 1.0$ are the maximum potentiation and depression amplitudes, and $\tau_{stp} = \tau_{std} = 20$ ms are the time constants determining the temporal window for plasticity. When presynaptic spikes preceded postsynaptic firing, synapses potentiated, which then strengthened inputs that reliably drove postsynaptic responses (Zannone et al., 2018). Conversely, a reversal in these dynamics led to depression, which would further eliminate connections that failed to contribute to output generation. Learning rates of 0.02 for unisensory pathways were chosen to balance rapid map formation against stability. Multisensory connections adapted more slowly to preserve established topography.

Recognising that NMDA receptors carry a larger fraction of charge than AMPA receptors at mature SC synapses, we maintained an AMPA:NMDA weight ratio of 1: 3 throughout training (Zhao et al., 2014). After every STDP update, the summed synaptic

strength of each unisensory to multisensory row was renormalised so that 25% of the weight resided in the fast AMPA component and 75% in the slow NMDA component, ensuring physiologically realistic but NMDA-dominated transmission while still allowing activity-dependent refinement of both receptors.

Unconstrained Hebbian plasticity has previously been shown to lead to instability through positive feedback loops (Wu et al., 2020). To counter such instability, we incorporated three homeostatic mechanisms operating at complementary timescales (Zenke et al., 2017). Fast Oja-type normalization, applied every external frame, maintained constant L2 norms for each neuron's afferent weight vector, implementing a form of hetero-synaptic competition (Witten et al., 2008). Medium-timescale inhibitory plasticity followed the Vogels-Abbott rule:

$$\Delta W_{inh} = \eta_i (\rho_{post} - \rho_{target}) s_{pre}$$

where ΔW_{inh} is the inhibitory weight change, $\eta_i = 0.01$ is the learning rate, ρ_{post} is the postsynaptic firing rate, $\rho_{target} = 5$ Hz is the target firing rate, and s_{pre} indicates presynaptic spikes. This homeostatic inhibition prevented runaway excitation while preserving stimulus selectivity. Finally, slow multiplicative scaling operated every 100 ms, gently nudging all synaptic weights toward a mean of 0.006 through soft rescaling (Wu et al., 2020). This three-faced homeostatic system reproduced the remarkable stability of SC responses despite ongoing plasticity.

Autism spectrum disorder model implementation. Converging evidence implicates specific synaptic alterations in the sensory processing differences observed in ASD (Chen et al., 2020; Robertson & Baron-Cohen, 2017; Siemann et al., 2017). We instantiated these findings through targeted parameter modifications that preserved overall network architecture while selectively altering synaptic transmission properties. NMDA conductance was reduced by 60% (from 0.05 to 0.02 normalized units), directly modelling the decreased NMDA receptor expression (Gandal et al., 2012), particularly of GluN2B subunits, reported in postmortem studies of autistic individuals. We then modelled for NMDA hyper-expression, by setting the NMDA conductance at 0.2, which is a fourfold increase from the training baseline (Marotta et al., 2020). These alterations affected the slow integrative component of excitatory transmission without affecting fast AMPA-mediated responses.

In addition to synaptic mechanisms, alterations in network and neuronal dynamics have also been implicated in ASD pathology. One such observation

frequently linked to differences in responses is altered E/I balance. We attempted to replicate the elevated E/I balance observed in ASD by proportionally reducing inhibitory tone, by reducing feedforward inhibition by 20% (Manyukhina et al., 2022). These changes weakened the competitive dynamics that normally sharpen spatial tuning and temporal precision. To test whether the temporal-widening effect of disinhibition depended on a single parameter choice, we performed a local sensitivity analysis around the two controller setpoints that determine the realized feedforward inhibitory gain across five independently trained checkpoints and re-estimating TBW width using the same fusion-curve fitting procedure described below.

Behavioural differences in ASD have also been linked to decreases in sensory adaptation (Lawson et al., 2015). To investigate the extent of change in responses induced by such a decrease, we reduced the neuronal sensory adaptation exhibited by the multisensory units. We did this by altering the Izhikevich variables of the neurons corresponding to recovery and reset. This combination of altered NMDA conductance, decreased inhibition, and lower sensory adaptation created a coherent model of ASD-related synaptic dysfunction grounded in empirical observations.

Behavioural Assays

All behavioural assays were performed on trained checkpoints, because the STDP-refined topographic alignment was necessary for stable spatial selectivity and biologically plausible multisensory binding curves. These assays were performed with long-term STDP/homeostatic updates frozen, while fast within-trial dynamics (AMPA short-term depression and intrinsic adaptation) remained active.

Temporal binding window. The temporal window over which the brain integrates multisensory information provides a fundamental measure of integration capacity (Marsicano et al., 2022). We quantified this window using a psychophysical fusion probability approach, systematically varying audiovisual onset asynchronies from -500 to +500 ms in 10 ms increments, where negative values indicated visual-leading and positive values indicated auditory-leading stimuli. For each offset, we presented five-frame (50 ms) stimulus pulses at 90° azimuth and recorded the integrated multisensory population response over a 100 ms window following the later stimulus onset. This integration period captured both immediate responses and NMDA-mediated sustained activity that bridges temporal gaps between inputs.

To derive fusion probabilities from neural responses, we transformed the integrated spike counts using a percentile-based normalization that preserved peak structure while mapping responses to a 0-1 probability scale. Specifically, we computed baseline and peak responses using the 10th and 90th percentiles respectively, then applied a soft-clipping transformation to maintain relative differences at high response levels. The resulting fusion probability curve was fit with a pedestal function, and the temporal binding window was defined as the range of asynchronies maintaining fusion probability above 50%. To permit quantitative comparison across conditions, TBW width was estimated separately for each checkpoint using the same 50% criterion psychometric fit applied to the fusion curves, and between-condition separation was summarized with Cohen's *d*. This metric directly relates to psychophysical measurements of perceived simultaneity in human subjects, where individuals with ASD consistently show broader temporal windows indicating less precise coincidence detection (Vogel et al., 2022).

Spatial binding window (SBW). While temporal alignment provides one binding cue, spatial congruence offers another critical constraint on multisensory integration. We assessed the network's spatial binding characteristics by presenting simultaneous audiovisual stimuli with systematically varied spatial disparities ranging from 0° to 80°. For each disparity level, we ran multiple trials with randomized absolute locations while maintaining the specified angular separation. The network's multisensory response pattern was analysed to determine whether it exhibited a single unified peak (indicating perceptual fusion) or multiple distinct peaks (indicating segregated processing).

We employed automated peak detection with Gaussian smoothing ($\sigma = 2$ neurons) to identify response maxima. A response was classified as "fused" if it contained a single peak above 20% of maximum amplitude, or if multiple peaks existed but the valley between them exceeded 60% of the lower peak height — indicating insufficient separation for independent representation. The proportion of fused responses across trials yielded a fusion probability curve as a function of spatial disparity. This curve was fit with a pedestal function to extract the spatial binding window width, defined as the disparity range maintaining 50% fusion probability. This measure captures the spatial principle of multisensory integration and its disruption in ASD, where enlarged spatial windows reflect reduced selectivity to spatial congruence.

Multisensory localization sensitivity. Multisensory integration should enhance perceptual sensitivity beyond what unisensory performance typically is, with the magnitude of enhancement depending on the relative reliability of each modality. We accounted for localization sensitivity by measuring the precision of spatial discrimination under audio-only, visual-only, and bimodal conditions. For each modality configuration, we generated sequences of 10-frame stimulus events at multiple azimuthal locations (5 trials per location) with fixed intensity (0.8) and added sensory noise ($\sigma = 0.1$).

Following each stimulus presentation, we decoded the perceived location using center-of-mass analysis of the multisensory population response. The signed localization error — the difference between decoded and true location — was computed for each trial. From the distribution of these errors, we calculated sensitivity as the inverse of the standard deviation:

$$\text{sensitivity} = \frac{1}{\sigma_{\text{error}}}$$

where σ_{error} represents the standard deviation of localization errors in degrees. Higher sensitivity values indicate more precise localization. We computed three sensitivity measures: s_A (auditory), s_V (visual), and s_B (bimodal). The multisensory enhancement of sensitivity was then quantified by comparing multisensory performance against the better unisensory modality. This measure revealed whether the network achieved optimal integration and whether this enhancement was preserved or diminished in the ASD model.

Cue reliability weighting. Optimal multisensory integration requires weighting sensory inputs according to their reliability — a computation fundamental to Bayesian theories of perception. We tested whether the network learned statistically optimal cue combination by independently manipulating the spatial uncertainty of auditory and visual inputs. Stimulus reliability was controlled by varying the spatial spread (σ) of Gaussian input profiles from 2° to 20° in 2° increments, with larger spreads representing less reliable spatial information.

For each combination of auditory and visual reliabilities, we presented spatially disparate stimuli with auditory input centered at 80° and visual input at 100° azimuth. To maintain equal total input despite varying spatial spreads, we applied gain normalization proportional to

$$\frac{\sigma_{ref}}{\sigma}$$

where $\sigma_{ref} = 2^\circ$ was the reference width. After 10 frames of stimulation, we decoded the perceived location from the integrated multisensory response and computed the visual weight as:

$$w_V = \frac{\text{decoded_location} - 80^\circ}{100^\circ - 80^\circ}$$

This weight represents the relative influence of visual information on the final percept, ranging from 0 (purely auditory) to 1 (purely visual). According to maximum likelihood estimation theory, the optimal visual weight should equal:

$$w_V^* = \frac{\sigma_A^2}{\sigma_A^2 + \sigma_V^2}$$

We compared empirically observed weights against these theoretical predictions across the full reliability matrix. The network's adherence to optimal cue combination was quantified through the mean absolute deviation between observed and predicted weights, as well as correlation analysis. This test revealed whether unsupervised learning through STDP could discover statistically optimal integration rules without explicit training on reliability-weighted tasks.

E/I ratio. Excitation and inhibition were quantified as model-internal synaptic current proxies rather than experimentally calibrated conductances. Each trained network was probed with a centred, synchronous audiovisual Gaussian pulse at 90° azimuth, and the synaptic current components falling on the multisensory excitatory population were recorded separately throughout a 125-ms evoked window. AMPA and NMDA currents were summed as excitation, whereas feedforward, recurrent, and lateral inhibitory currents were summed as inhibition; these signals were then integrated over time to produce evoked excitatory and inhibitory charge in arbitrary model units. The E/I balance metric was defined as evoked excitatory charge divided by evoked inhibitory charge and then averaged across independently trained network instantiations. Accordingly, Fig. 1b is intended only to place the model in an excitation- versus inhibition-dominant operating regime and to provide qualitative context, rather than a direct fit to experimentally measured conductances or currents.

Inverse effectiveness. The principle of inverse effectiveness, whereby multisensory enhancement is greatest when unisensory responses are weakest, represents a fundamental computation that optimizes information gain under uncertainty. We

quantified this phenomenon by measuring multisensory spike-count enhancement across stimulus intensities (0.05 to 1.6 arbitrary units). For each intensity and modality combination, we presented 10-frame stimulus pulses at 90° azimuth and recorded the integrated multisensory spike count over 20 frames.

The multisensory enhancement index (MEI) was computed as:

$$\text{MEI} = \frac{R_{AV} - \max(R_A, R_V)}{\max(R_A, R_V)}$$

where R_{AV} represents the multisensory response and $\max(R_A, R_V)$ represents the stronger unisensory response at each intensity. By plotting MEI against stimulus intensity on a logarithmic scale, we assessed whether the network exhibited the characteristic negative relationship indicative of inverse effectiveness. This nonlinear relationship would indicate that multisensory integration provides maximal benefit when sensory signals are weakest and most ambiguous, precisely when enhanced detection would be most behaviourally advantageous.

Distance-dependent noise correlations. We quantified the spatial scale of shared trial-to-trial variability by computing pairwise spike-count noise correlations between multisensory neurons and binning them by separation along the azimuthal map. For each trained model, we ran repeated, identical bimodal (simultaneous A+V) trials at a fixed location and accumulated per-trial multisensory spike counts over the stimulus sequence. Counts were z-scored across trials for each neuron and noise correlation was defined as the Pearson correlation coefficient of the resulting per-trial spike counts for each neuron pair, a standard measure for r_{sc} (i.e., spike-count correlation). Neuron positions were mapped to azimuth on a 0–180° ring; pairwise separation was the minimal circular distance $d = \min(|\Delta|, 180 - |\Delta|)$ restricted to unique distances $d \in [0, 90^\circ]$.

To prevent inflation of the near-zero bin by self-pairs and clone-like receptive fields, we excluded all pairs with d lesser than one neuron pitch (degrees per neuron). Remaining pairs were grouped into 1-pitch bins using half-open edges $(a, b]$, beginning at $1 \times \text{pitch}$, and we computed the mean correlation within each bin. This produced a per-model curve that was then aggregated across $n = 10$ independently trained networks to obtain the pooled mean $\pm$ SEM at each distance; for visualization only, the pooled trace was lightly smoothed with a Gaussian kernel ($\sigma = 1.2$ bins) using reflect boundaries to avoid edge artifacts.

Response latency analysis. Multisensory integration in the SC not only enhances response magnitude but also accelerates reaction times, a phenomenon critical for rapid orienting behaviours. We measured first-spike latencies by presenting 10-frame stimulus pulses under audio-only, visual-only, and bimodal conditions, then detecting the first multisensory spike that exceeded baseline activity. The network was run for 40 frames total to ensure capture of delayed responses, with latency recorded as the time from stimulus onset to first spike in milliseconds.

For each network, we computed the mean unisensory latency (averaging audio and visual conditions) and compared it to the multisensory latency. The latency reduction

$$\Delta\text{latency} = \frac{L_A + L_V}{2} - L_B$$

quantified the temporal advantage conferred by multisensory integration. This measure allowed us to compare our latency measurements to behavioural studies showing reduced multisensory facilitation of reaction times in individuals with ASD, potentially reflecting weakened temporal summation mechanisms.

Data and code availability. The data and code used in the current study are available [here](#).

RESULTS

Model validation and characterisation. We developed a two-layer SNN to investigate how synaptic alterations affect multisensory processing in the SC. The architecture incorporated separate auditory and visual input layers that projected to a multisensory integration layer through both excitatory and inhibitory pathways (**Fig. 1a**). Following 80 epochs of unsupervised training with pre-set stimulus statistics (60% bimodal, 20% audio-only, 20% visual-only events), we measured the ratio of inhibitory to excitatory currents in response to centered bimodal stimuli. The network maintained an E/I ratio of 1.04 ± 0.04 across ten independent instantiations (**Fig. 1b**), consistent with the approximately 1:1 balance documented in cortical and subcortical circuits. This balance emerged despite implementing a 30% proportion of inhibitory neurons and three distinct inhibitory mechanisms (feedforward, recurrent, and lateral), suggesting that STDP mechanisms successfully regulated synaptic weights to prevent unstable dynamics.

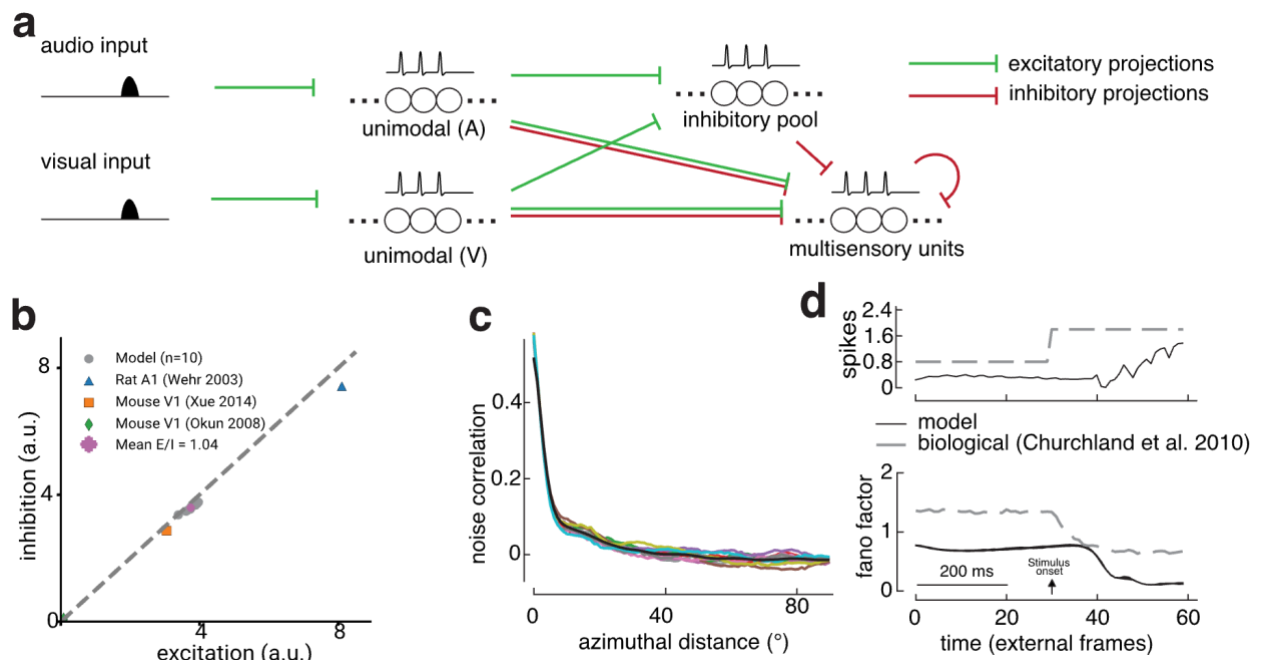


Figure 1. Network architecture and validation of key physiological properties. **a**) Two-layer spiking neural network with auditory and visual input layers projecting to multisensory (MSI) layers through excitatory (green) and inhibitory (red) connections. **b**) Evoked excitatory and inhibitory synaptic charge in the model, computed by separately integrating AMPA+NMDA excitation and feedforward, recurrent, and lateral inhibitory currents over the 125-ms response window to a synchronous central audiovisual stimulus (a.u., arbitrary model units); grey points show individual network instantiations ($n = 10$), the magenta point shows the mean E/I ratio, and selected published reference values are shown for qualitative context. **c**) Spike-count noise correlation as a function of azimuthal distance between neuron pairs in the multisensory layer. The coloured lines represent noise correlation of separate network instances. The black line represents mean across instances. **d**) Fano factor dynamics showing spike count variability (top) and fano factor (bottom) following stimulus onset; black line shows the model results, gray dashed line shows biological reference data from Churchland et al. (2010).

To validate the mesoscale coupling in the multisensory layer, we quantified spike-count “noise” correlations (r_{sc}) across repeated presentations of identical stimuli and binned neuron pairs by their separation on the azimuth map (0–80°). Correlations were strongest for nearby pairs and declined monotonically with azimuthal distance, approaching near-zero at large separations (**Fig. 1c**). This spatial fall-off is the expected footprint of local shared input in sensory populations and mirrors the well-established distance-dependence of correlated variability in primate cortex and, specifically for SC, the observation that r_{sc} depends on pairwise proximity and functional class. Together with the near-balanced E/I ratio, this shows the trained network learns realistic local coupling without imposing spurious long-range common noise.

We examined trial-to-trial variability by computing the Fano factor (variance/mean) of spike counts in 10 ms bins. Under the weak background validation regime, the network exhibited a Fano factor of 0.73 ± 0.05 , which decreased to 0.21 ± 0.03 within 100 ms following stimulus presentation (**Fig. 1d**). This reduction occurred concurrently with an increase in mean firing rate from 0.4 to 1.6 spikes per bin. The temporal dynamics and trend of this variability reduction paralleled recordings from macaque middle temporal (MT) cortex, where Fano factors decrease from ~ 1.5 to ~ 0.7 following motion stimulus onset. Because prestimulus input varied independently across trials whereas the stimulus drive was shared, stimulus onset itself quenched trial-to-trial variability; comparable FF suppression persisted after removal of inhibitory pathways in control simulations, indicating that inhibition may contribute to recurrent stabilization but is not necessary for the effect, in this model.

Emergent multisensory integration properties. Having established that the network maintained balanced E/I and correlated noise activity, we next examined whether it reproduced characteristic signatures of multisensory integration observed in SC neurons and psychophysical studies. Consistent with this, freshly instantiated networks failed to develop aligned receptive fields and did not reproduce the selective temporal or spatial binding profiles observed after training. We evaluated five fundamental properties: enhanced localization sensitivity, spatiotemporal binding

windows, optimal cue weighting, accelerated response latencies, and inverse effectiveness.

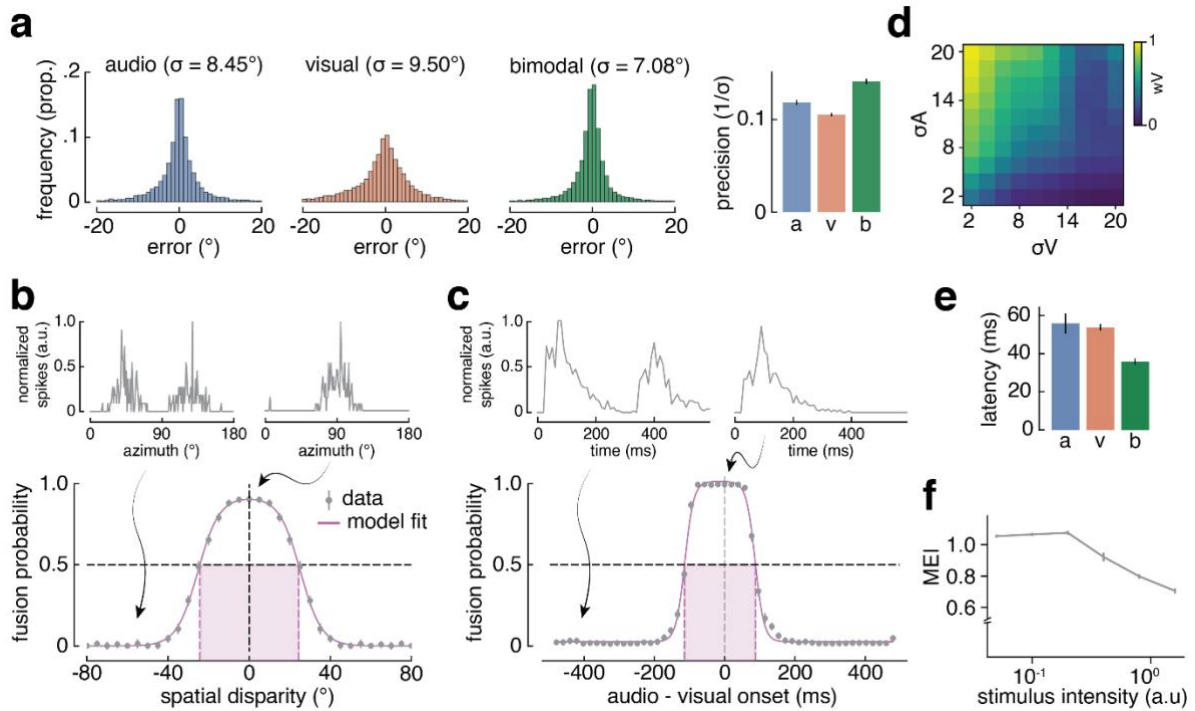


Figure 2. Multisensory integration properties in the trained network. **a)** Frequency distributions of signed localization errors for auditory ($\sigma=8.47^\circ$), visual ($\sigma=9.59^\circ$), and bimodal ($\sigma=7.08^\circ$) stimulus conditions (left histograms) with corresponding mean precision values ($1/\sigma$) averaged across ten network instantiations (right bar plot). **b)** Spatial binding window analysis. Upper panel: example multisensory population spike activity across azimuthal positions ($0-180^\circ$) showing fused and segregated responses at different audiovisual spatial disparities. Lower panel: Fusion probability as a function of spatial disparity between audiovisual stimuli, with magenta shaded region indicating the binding window where fusion exceeds 50%. **c)** Temporal binding window analysis. Upper panel: multisensory population spike activity over time showing responses to temporally aligned and asynchronous audiovisual stimuli. Lower panel: Fusion probability as a function of audiovisual onset asynchrony (negative values: visual-leading, positive: auditory-leading), with shaded region marking the temporal binding window. **d)** Heat map displaying visual weight (wV) in multisensory localization as a function of auditory (σA) and visual (σV) stimulus reliability, where larger σ values represent less reliable spatial information. **e)** Mean first-spike latencies for auditory, visual, and bimodal stimulus conditions. **f)** Multisensory enhancement index (MEI) plotted against stimulus intensity on a logarithmic scale. Error bars indicate $\pm$ SEM across ten independent network instantiations. Gray dots in panels b and c represent mean trial measurements across ten independent network instantiations; solid lines show fitted curves.

The network demonstrated enhanced spatial localization when both sensory modalities were present compared to either alone (**Fig. 2a**). Analysis of signed localization errors revealed that auditory-only trials produced a broad error distribution ($\sigma = 8.47^\circ$), while visual-only trials showed comparable but slightly lower precision ($\sigma = 9.59^\circ$). Bimodal stimulation yielded the highest localization sensitivity ($\sigma = 7.08^\circ$), corresponding to a 19% improvement over the better unisensory condition. This enhancement pattern mirrors the multisensory integration effects observed in SC neurons of behaving animals, where responses to bimodal stimulation typically

outperform unimodal spatial discrimination under optimal stimulus conditions (Stanford et al., 2005). Such multisensory enhancement is also observed across human behavioural responses to spatial discrimination tasks (Alais & Burr, 2004; Battaglia et al., 2003; Binda et al., 2007). The error distributions for all three conditions approximated Gaussian profiles centred around zero, indicating unbiased spatial representations.

The spatial window over which the network integrated multisensory information revealed distance dependent constraints on binding (**Fig. 2b**). We presented simultaneous audiovisual stimuli with varying angular disparities and measured the probability of observing a unified or segregated population response (i.e., either one or two distinct peaks in activity in the multisensory layer as shown in the upper panel of **Fig. 2b**). The network maintained fusion probability above 50% for disparities up to $\pm 24^\circ$, beyond which fusion declined to baseline levels by $\pm 40^\circ$. This spatial binding window corresponds to the receptive field overlap required for multisensory enhancement in SC neurons. By contrast, the temporal binding window, assessed by varying audiovisual onset asynchronies, exhibited marked asymmetry in integration capacity (**Fig. 2c**). Fusion probability remained above 50% for asynchronies ranging from -121 ms (visual leading) to +93 ms (auditory leading), with peak integration at slight visual-lead offsets. This asymmetry aligns with the differential conduction delays in the network architecture: visual signals required 26 ms to reach multisensory neurons while auditory signals required only 10 ms, creating a natural bias toward visual-preceding-auditory sequences that may compensate for faster auditory processing in natural environments. A step-size sensitivity analysis on a representative trained checkpoint, halving the internal step size to $dt = 0.05$ ms while doubling the number of substeps to preserve the 10 ms frame duration, yielded a qualitatively similar TBW profile (normalized curve correlation $r=0.94$, normalized RMSE = 0.10; **Supplementary Fig. 1**)

To assess whether the network learned appropriate cue combination strategies, we manipulated the spatial reliability of auditory and visual inputs by varying their spatial spread from $\sigma = 2^\circ$ to 20° . When both modalities had equal reliability, the network assigned similar weights to each input ($w_V = 0.51 \pm 0.03$). As we decreased visual reliability relative to auditory, visual weight decreased systematically to 0.04 ± 0.04 when $\sigma_V = 20^\circ$ and $\sigma_A = 2^\circ$ (**Fig. 2d**). This pattern suggests that the network learned to dynamically weight sensory inputs based on their reliability, with more

reliable modalities receiving stronger weights. The empirical weights showed a strong correlation with maximum likelihood estimation predictions ($R^2 = 0.71$), with a mean absolute error of 0.13, indicating that exposure to correlated multisensory events during training was sufficient for the network to learn reliability-based integration. While not numerically optimal, this weighting strategy approximates the principles of statistically optimal integration, allowing the network to rely more heavily on the sensory channel providing higher quality information (Alais & Burr, 2004; Battaglia et al., 2003).

Response timing analysis demonstrated that multisensory stimulation accelerated neural responses relative to unisensory conditions (**Fig. 2e**). Mean first-spike latencies were 56 ± 5.2 ms for auditory, 54 ± 1.6 ms for visual, and 36 ± 1.6 ms for bimodal stimulation across network instantiations. The 19 ms reduction in multisensory latency represents a 35% acceleration compared to the mean unisensory latency, an effect size comparable to reaction time benefits observed in multisensory detection tasks. This latency reduction emerged from the convergence of subthreshold inputs that, when combined, reached spike threshold earlier than either modality alone. We also tested for the principle of inverse effectiveness, whereby multisensory enhancement is greatest for weak stimuli, manifested in the network's responses across stimulus intensities (**Fig. 2f**). The multisensory enhancement index (MEI) decreased from 1.1 at the lowest intensity (0.05 a.u.) to 0.78 at the highest (1.6 a.u.), broadly following a power-law relationship on logarithmic axes. This inverse relationship ensures that multisensory integration provides maximal benefit when unisensory signals are least reliable, a computational strategy that optimizes behaviour under conditions of sensory uncertainty.

Reduced neuronal adaptation. Diminished sensory adaptation has emerged as a potential mechanism underlying atypical sensory processing in autism, with individuals showing reduced habituation to repeated stimuli across multiple sensory modalities (Dwyer et al., 2023; Puts et al., 2014). To probe how diminished spike-frequency adaptation shapes multisensory behaviour, we re-parameterised multisensory neurons from regular-spiking ($a = 0.02$, $d = 8$) to near tonic ($a = 0.0001$, $d = 0.01$, $b = 0.2$, $c = -60$ mV), greatly reducing the after-hyperpolarising current that curtails sustained firing. This manipulation allowed us to examine whether altered intrinsic

excitability could contribute to the characteristic sensory integration differences observed in autism spectrum disorder (Millin et al., 2018).

The spatial binding window expanded significantly following adaptation reduction (**Fig. 3a**). While the control networks maintained fusion probability above 50% for audiovisual disparities within $\pm 24^\circ$, perturbed networks extended this range to $\pm 29^\circ$, a 21% increase in spatial tolerance. Additionally, the fusion probability curve developed a flat-top plateau, maintaining higher fusion across the central $\pm 20^\circ$ range rather than the narrower window observed in controls. This resembles reports that individuals with ASD often show more implicit integration at large spatial disparities (Noel et al., 2022). The distortion of graded spatial selectivity we see in our perturbed networks may suggest that reduced adaptation prevents the competitive dynamics necessary for precise spatial discrimination, allowing disparate inputs to drive sustained multisensory responses (Stein et al., 2014).

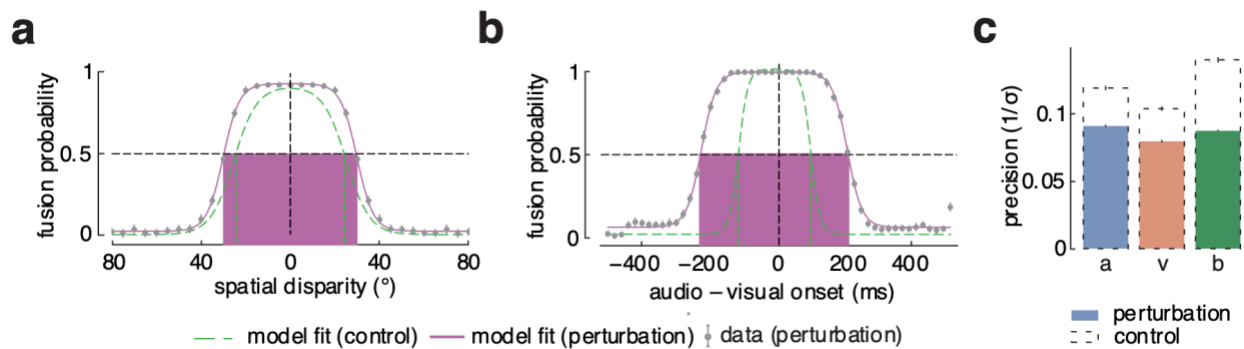


Figure 3. Effects of reduced neuronal adaptation on multisensory integration. **a)** Fusion probability as a function of spatial disparity for control (green dashed line) and reduced adaptation (magenta solid line) networks. **b)** Fusion probability as a function of audio-visual onset asynchrony for control and perturbed networks. **c)** Localization precision ($1/\sigma$) for auditory, visual, and bimodal conditions; coloured bars show perturbed networks with black dashed lines indicating control values. Vertical green dashed lines and shaded regions in **(a, b)** indicate where fusion probability exceeds 50% for control and perturbed networks, respectively; negative asynchronies in **(b)** indicate visual-leading stimuli. Error bars indicate $\pm$ SEM across ten network instantiations.

Temporal integration constraints showed comparable alterations. Post manipulation, fusion tolerance to onset asynchrony increased dramatically, expanding from -121 to +93 ms (215 ms total) in controls to -229 to +201 ms (431 ms total) post manipulation, with the bulk of the expansion occurring on the visual lead side (**Fig. 3b**). Both the magnitude and asymmetry of this temporal expansion aligns with behavioural studies in ASD, where individuals consistently show temporal binding windows ~ 400 ms on flash-beep simultaneity judgments and demonstrate particular difficulty rejecting visual first sequences (de Boer-Schellekens, Keetels, et al., 2013;

Foss-Feig et al., 2010; Stevenson, Siemann, et al., 2014). Within the network architecture, reduced adaptation (and thus uncontrolled spiking) prolongs NMDA-mediated depolarization, potentially extending the window during which late arriving auditory signals can trigger multisensory enhancement. This mechanism is consistent with evidence of altered NMDA receptor expression and function in ASD, particularly in subcortical structures involved in sensory integration (Won et al., 2012).

The expanded binding windows exhibited by the modified network were accompanied by degraded localization performance (**Fig. 3c**). Sensitivity ($1/\sigma$) decreased by approximately 23% for unimodal and 37% for bimodal stimulation, relative to controls. The typical multisensory benefit (whereby sensitivity to bimodal stimulation exceeds that to both unimodal stimuli) was substantially diminished in adaptation-reduced networks. Control networks showed 19% enhancement for multisensory compared to (best) unisensory performance, while perturbed networks showed a 4.6% diminution. These findings suggest that the mechanism contributing to broadening of the integration windows may also increase response variability, degrading the signal-to-noise ratio that typically enhances multisensory localization. Our results parallel the contrasting combination of enhanced binding yet reduced behavioural benefit from multisensory integration commonly observed in ASD (Noel et al., 2022; Ostrolenk et al., 2019).

Reducing spike-frequency adaptation fundamentally altered network dynamics by allowing multisensory neurons to maintain prolonged depolarisation, thus extending the window over which incoming sensory signals could summate effectively. Under normal conditions, adaptation currents quickly repolarise neurons after firing, sharply restricting the temporal window for multisensory integration. Without this rapid repolarisation, NMDA receptors remained partially unblocked for an extended duration (17% compared to 10%), increasing the cumulative NMDA-mediated excitation, which resultantly increased the share of NMDA charge, relative to AMPA (by ~40% on average). This sustained excitation effectively overwhelmed inhibition that, although recruited at the same latency as control conditions (~2 ms after excitation), was insufficient to rapidly repolarise multisensory neurons. At the same time, the prolonged NMDA currents and excitation also affected neighbouring neurons, manifesting as persistent local recurrent excitation that overpowered the inhibitory surround (the increased NMDA current into the neurons decreased the depth of the firing activity valley between two adjacent spiking neural populations and resulted in increased

fusion across spatial disparities, despite surround inhibition increasing from 1.5 to 2.6 following reduced adaptation). Consequently, the prolonged excitation resulted in the expanded temporal and spatial binding windows, coupled with a reduction in localisation precision. This pattern of indiscriminate fusion and reduced precision closely mirrors sensory processing anomalies frequently reported in ASD populations (Noel et al., 2022). Such findings support the notion that diminished neuronal adaptation may underlie core aspects of multisensory integration deficits observed in autism.

Reduced feedforward inhibition. Disrupted excitation-inhibition balance represents one of the most widely documented circuit alterations in autism spectrum disorder, with evidence spanning from genetic mutations affecting GABAergic signalling to altered inhibitory neuron density in postmortem tissue (Nelson & Valakh, 2015; Rubenstein & Merzenich, 2003). We next sought to examine how altered E/I balance affects multisensory integration and to that extent, we scaled feedforward GABAergic inhibition from unisensory to multisensory layers to 80% of their baseline values. This manipulation increased the epoch-integrated E/I charge ratio from approximately 1.1 in control networks to 3.0 in perturbed networks, mimicking the reduction in cortical inhibition hypothesized in ASD (Antoine et al., 2019; Rubenstein & Merzenich, 2003; Sohal & Rubenstein, 2019). This intervention produced integration deficits similar to those observed with reduced adaptation but differing in magnitude, which may reflect the way different circuit perturbations generate heterogeneous sensory phenotypes within the autism spectrum (Hahamy et al., 2015; Lombardo et al., 2019; Sohal & Rubenstein, 2019; Zerbi et al., 2021).

The spatial binding window expanded following inhibition reduction, with the 50% fusion threshold expanding from $\pm 24^\circ$ to $\pm 27^\circ$, a 15% increase in spatial tolerance (**Fig. 4a**). By analogy with the adaptation manipulation, where reduced adaptation broadened the spatial binding window via greater NMDA conductance and more persistent excitation (**Fig. 3a**), a simple E/I account would predict that feedforward disinhibition should likewise widen, not narrow, the spatial binding window. This is indeed what is observed here, where reduced feedforward inhibition broadened spatial tuning.

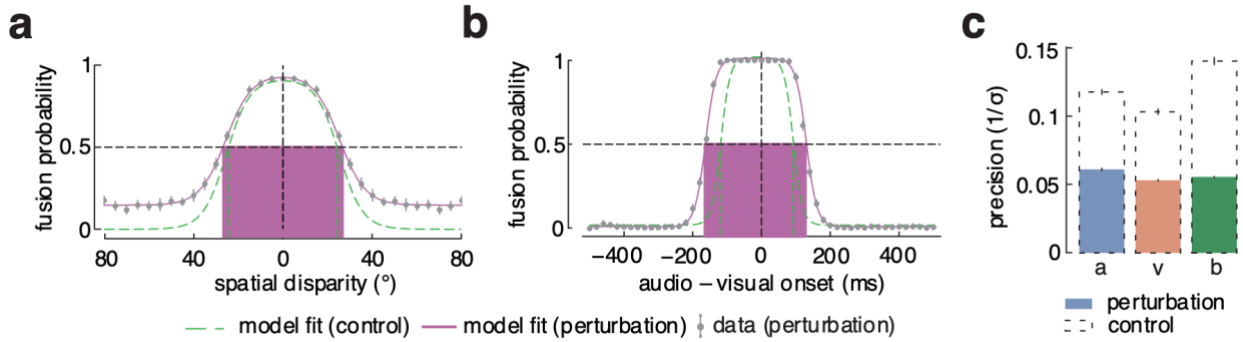


Figure 4. Effects of reduced feedforward inhibition on multisensory integration. a) Fusion probability as a function of spatial disparity for control (green dashed line) and reduced feedforward inhibition (magenta solid line) networks. **b)** Fusion probability as a function of audio-visual onset asynchrony for control and perturbed networks. **c)** Localization precision ($1/\sigma$) for auditory, visual, and bimodal conditions; coloured bars show perturbed networks with black dashed lines indicating control values. Vertical green dashed lines and shaded regions in (a, b) indicate where fusion probability exceeds 50% for control and perturbed networks, respectively; negative asynchronies in (b) indicate visual-leading stimuli. Error bars indicate $\pm$ SEM across ten network instantiations.

Temporal binding also underwent a typical expansion following disinhibition, expanding from -121 to +93 ms (215 ms total) in controls to -163 to +131 ms (294 ms total) following disinhibition (**Fig. 4b**), a 36% expansion of the temporal window. The temporal expansion coincided with prolonged NMDA-mediated depolarization, as indicated by increased average Mg^{2+} gate relief (12.5% compared with 10.4% in controls). Under these conditions, sensory inputs separated by up to ~300 ms could potentially summate effectively, particularly when visual stimuli preceded auditory ones. This finding was expected, given that feedforward inhibition typically narrows the temporal integration window (Gabernet et al., 2005; Pouille & Scanziani, 2001). Extended audiovisual temporal binding windows are a robust sensory marker of ASD, appearing across flash-beep, speech, and complex stimulus paradigms (Stevenson, Siemann, et al., 2014; Zhou et al., 2018). A local sensitivity analysis across ten independently trained checkpoints showed that this temporal-widening effect was not restricted to a single tuned setting: relative to control networks (TBW = 212.4 ± 4.9 ms), reducing the feedforward-inhibition controller setpoints to 0.8 widened TBW to 291.0 ± 6.0 ms, and reducing them to 0.5 widened it further to 300.8 ± 5.7 ms, with widening observed in all 10/10 checkpoints in both conditions.

Concurrent with spatiotemporal binding alterations, reduced feedforward inhibition led to less precise localization across all conditions, with sensitivity declining by 48% for auditory, 48% for visual, and 60% for bimodal stimuli (**Fig. 4c**). Control networks showed 19% enhancement for multisensory compared to (best) unisensory performance, while perturbed networks showed a 10% diminution. The

disproportionate impact on multisensory performance indicates that elevated excitation not only increased baseline response variability but specifically disrupted the uncertainty reduction mechanisms that might otherwise enhance multisensory precision (Dinstein et al., 2012). Such loss of spatial acuity parallels experimental findings where GABA receptor blockade or optogenetic silencing of interneurons in the superior colliculus produces comparable degradation of spatial discrimination and multisensory gain (Binns & Salt, 1997; Wang et al., 2020).

The dissociation observed in our networks (expanded spatiotemporal binding windows coupled with reduced precision) indicates that E/I imbalance differentially affects distinct aspects of integration (Pouille & Scanziani, 2001; Wilson et al., 2012). Our findings indicate that while reduced adaptation produces uniformly expanded binding across spatial and temporal domains, an elevated E/I ratio generates similar effects but varying in magnitude. Such mechanistic diversity could potentially account for the variable sensory profiles observed clinically (Noel et al., 2022; Weiland et al., 2023; Zhou et al., 2018), though direct experimental validation would be necessary to establish whether similar differences occur in biological systems.

NMDA alterations. Exome sequencing has identified mutations in NMDA receptor subunit genes (GRIN1, GRIN2A, GRIN2B) in individuals with autism spectrum disorder, with both receptor hypofunction and hyperfunction implicated in different contexts (Cha et al., 2025; De Rubeis et al., 2014; O’Roak et al., 2011). To investigate how NMDA receptor function shapes multisensory integration, we systematically altered NMDA conductance in both directions, reducing NMDA expression from 0.05 to 0.02 (60% reduction) and increasing it to 0.2 (fourfold increase). These manipulations allowed us to examine whether NMDA receptors primarily contribute to the temporal precision of coincidence detection or more broadly regulate the spatiotemporal constraints of multisensory binding.

Reduced NMDA conductance produced consistent narrowing of integration windows across both spatial and temporal domains (**Fig. 5a, b**). The spatial binding window contracted from $\pm 24^\circ$ to approximately $\pm 12^\circ$, with fusion probability declining more steeply as audiovisual disparities increased. This narrowed spatial tolerance may reflect limited capacity for sustained depolarization when NMDA-mediated currents are weakened (Ewald et al., 2008; Mayer et al., 1984), potentially restricting the spatial extent over which convergent inputs can fuse. Temporal integration showed

minimal constriction, with the binding window shrinking from -121 to +93 ms (215 ms total) in controls to -107 to +78 ms (185 ms total) following NMDA reduction. The asymmetric nature of this minimal narrowing, with evident impact only on audio leading asynchronies, could result from insufficient slow integrative currents to bridge the temporal gap between early auditory and delayed visual inputs (Drotos et al., 2024). Localization sensitivity declined across all conditions, with multisensory precision showing relatively more degradation (30% degradation across unisensory sensitivities and 33% for multisensory; **Fig. 5c**). The reduced multisensory enhancement (18% multisensory enhancement for control compared to 10% enhancement following manipulation) supports previous claims that adequate NMDA function is key for deriving signal processing benefits from combining sensory information (Lebedeva et al., 2019; Li & Gulledge, 2021).

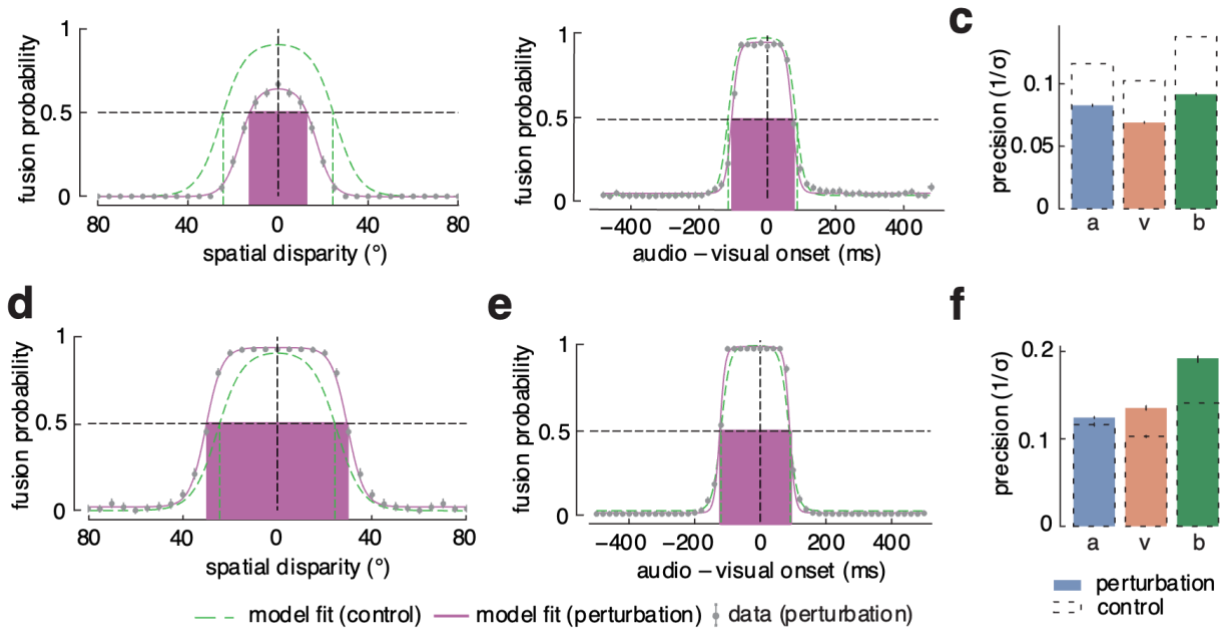


Figure 5. Bidirectional effects of NMDA receptor conductance alterations on multisensory integration. **a-c)** Reduced NMDA conductance (60% reduction): **a)** Fusion probability as a function of spatial disparity for control (green dashed) and NMDA-reduced networks (purple solid). **b)** Fusion probability as a function of audio-visual onset asynchrony; negative values indicate visual-leading, positive indicate auditory-leading stimuli. **c)** Localization precision ($1/\sigma$) for auditory, visual, and bimodal conditions; coloured bars represent perturbed networks with control values shown as black dashed lines. **d-f)** Increased NMDA conductance (fourfold increase): **d)** Spatial fusion probability, **e)** Temporal fusion probability, and **f)** Localization precision, following the same format as panels a-c. Vertical green dashed lines and shaded regions in (**a**, **b**, **d**, **e**) indicate where fusion probability exceeds 50% for control and perturbed networks, respectively; negative asynchronies in (**b**, **e**) indicate visual-leading stimuli. Error bars indicate $\pm$ SEM across ten network instantiations.

Increased NMDA conductance led to a different pattern of alterations that revealed comparable effects on spatial and temporal processing (**Fig. 5d, e**). The

spatial binding window underwent minimal yet significant expansion going from $\pm 24^\circ$ to approximately $\pm 30^\circ$, with fusion probability approaching unity across the central $\pm 20^\circ$ range, above the fusion probability observed in controls. This plateaued fusion profile suggests that enhanced NMDA currents could create overly permissive spatial integration, potentially allowing misaligned inputs to drive fused percepts (Rhodes, 2006). The temporal binding in contrast showed little difference from -121 to +93 ms (215 ms total) in controls to -123 to +92 ms (214 ms total) following an increase in NMDA conductance, while maintaining the same asymmetric profile as control networks (Drotos et al., 2024). Localization sensitivity showed improvements across all stimulus conditions (1% enhancement across audio sensitivity, 25% increase across visual and 33% increase across bimodal stimuli), with the multisensory benefit in localization accuracy showing enhancements following manipulation (19% multisensory enhancement in controls compared to 41% enhancement following perturbation). The enhanced NMDA conductance likely improved localization accuracy by stabilizing synaptic integration and reducing location-dependent variability in neuronal responses (Li & Gulledge, 2021). This would effectively increase the fidelity of EPSP-spike coupling across spatial domains despite the more permissive binding window, thereby creating more reliable spatial representations even when fusion probability increased.

The bidirectional effects seen during NMDA manipulation here suggests that these receptors play a complex role in multisensory processing. In our model, NMDA receptors appear to provide sustained depolarization that facilitates spatial integration across topographic maps, with reduced NMDA limiting effective spatial summation (Ewald et al., 2008; Monyer et al., 1994) and elevated NMDA potentially causing indiscriminate binding of spatially disparate inputs. The preservation of multisensory enhancement with increased NMDA contrasted with its elimination following NMDA reduction, suggests that while adequate NMDA function may be necessary for optimal integration, the system appears more tolerant of excess than deficiency in our network architecture.

DISCUSSION

Mechanistic heterogeneity in ASD multisensory processing. Here for the first time, a biologically constrained SNN model of the SC was deployed to investigate how particular synaptic and cellular perturbations give rise to deficits in multisensory integration in ASD. Our systematic manipulation of neuronal adaptation, E/I balance and NMDA receptor conductance showed that each perturbation produced alterations in spatiotemporal binding. Reduced adaptation yielded expansions in binding windows, diminished feedforward inhibition generated comparable effects that differed in magnitude and NMDA receptor alterations produced bidirectional, magnitude-dependent changes. Our findings show that the neural mechanisms tested here can lead to the enlarged temporal binding windows observed in ASD populations (Foss-Feig et al., 2010). They also produce distinct effects on spatial integration (Noel et al., 2022) and localization precision (Fujihira et al., 2022; Visser et al., 2013). This contrast in results may account for the variability in symptoms characteristic of autism (Lombardo et al., 2019).

The differences observed across our three perturbation conditions imply that a set of distinct mechanisms may give rise to the behavioural deficits associated with ASD. Reduction of adaptation allowed for long active times for the neurons, extending their temporal windows of integration by sustaining NMDA receptor activation. Disinhibition led to similar effects on binding windows, although the magnitude of change varied. Modifications of NMDA receptor had graded effects that scaled with the level of conductance, which bidirectionally impacted spatiotemporal processing. These mechanisms reflect a frequently reported clinical observation: individuals with ASD can show similar extended spatiotemporal binding deficits yet differ markedly in other aspects of sensory processing. Instead of viewing ASD as deviations along a single axis of dysfunction, our results suggest that different combinations of cellular or synaptic modifications could lead to the same behavioural responses, while varying in their computational pathologies. Understanding these differences in mechanistic terms requires an examination of how each perturbation alters the fundamental operations of the multisensory neurons.

Perturbation-specific computational mechanisms. A reduction in neural adaptation led to disruptions in multisensory integration, extending spatiotemporal windows by significant margins. When we converted multisensory neurons from regular-spiking to

near-tonic firing patterns, a reduction in spike frequency adaptation stopped the polarisation that normally followed stimulus driven activity, maintaining NMDA receptors in a partially unblocked state for extended periods (the average gate relief increased from 10% to 17%; Zhang-Hooks et al., 2016). This sustained depolarization created a computational state where neurons could no longer separate when one stimulus ended and another began, nor maintain precise spatial boundaries between adjacent inputs, explaining why both temporal and spatial windows expanded proportionally (Kwakye et al., 2011). The mechanism resembles the "sticky attention" phenomenon observed clinically (Dwyer et al., 2024; Zwaigenbaum et al., 2005), where individuals with ASD show difficulty disengaging from stimuli, suggesting that adaptation deficits might underlie the automatic orienting impairments characteristic of the condition (Gonçalves & Monteiro, 2023). A reduction in adaptation stands out in comparison to the other perturbations discussed here, because while they affect different aspects of integration in an asymmetric fashion, disrupting adaptation can compromise the basic ability of a neuron to reset between stimuli (Takarae & Sweeney, 2017).

While adaptation altered intrinsic reset, feed-forward disinhibition separates where and for how long inputs are bound. Weakening the unimodal-to-multisensory GABAergic projection led to expansions in binding windows (Kwakye et al., 2011; Lee et al., 2020; Robertson et al., 2013). This distinction arose through measurable circuit reorganization. With a decrease in the strength of early inhibitory gating, aligned audiovisual inputs generated stronger central responses that also led to an increase in the strength of lateral inhibition (feedforward inhibition decreased by 20% while lateral inhibition increased by 5.7%), but this increase in lateral inhibition was not enough to compensate for the loss of feedforward inhibition and as a result the spatial tuning of those neurons widened (Schallmo et al., 2020).. Simultaneously, loss of early GABA shunting allowed prolonged NMDA accumulation, with the NMDA/AMPA ratio rising 27% and extending temporal integration windows. The recurrent inhibitory loops remained stable throughout, as they were not directly affected by changes to feedforward inhibition. These findings reveal how the strength and timing of inhibition determines if and how audiovisual inputs bind across space and time (Kwakye et al., 2011; Robertson et al., 2013). Such inhibition mediated effects may explain why some individuals with ASD show conflicting spatial attention windows alongside difficulty

filtering stimuli separated in time (Crasta et al., 2021; Robertson et al., 2013; Schallmo et al., 2020).

By contrast, tuning NMDA currents acted as a molecular time-constant that bidirectionally regulated integration windows (Paoletti et al., 2013). Reducing NMDA conductance by 60% narrowed both spatial ($\pm 24^\circ$ to $\pm 12^\circ$) and temporal (215 to 185 ms) binding windows, while fourfold enhancement expanded spatial ($\pm 24^\circ$ to $\pm 30^\circ$) with barely any temporal affect (215 to 216 ms) windows with improved localization precision (Binns & Salt, 1996). This partly bidirectional relationship resembles the developmental GluN2B→GluN2A subunit switch, where early GluN2B-enriched receptors cause longer decay times and broader integration windows that narrow as GluN2A expression increases (Bellone & Nicoll, 2007; Brigman et al., 2010). Multiple GRIN2B variants identified in ASD cohorts alter receptor kinetics in opposing directions, some reducing channel open time while others enhance conductance (Fedele et al., 2018; Sabo et al., 2023). This could explain why the temporal binding window of some individuals with ASD is expanded, while for others it is contracted (de Boer-Schellekens, Keetels, et al., 2013; Kwakye et al., 2011). Our model suggests these opposing clinical observations may reflect a continuous spectrum of NMDA function rather than distinct pathophysiological mechanisms (Zoodsma et al., 2022). This model shows an increase in multisensory enhancement with increased NMDA but an elimination of enhancement upon lowering NMDA conductance, which goes on to imply that integration benefits require a minimum threshold of slow synaptic current, below which coincidence detection fails regardless of fast AMPA-mediated transmission (Binns & Salt, 1996).

The responses demonstrated by our model following NMDA manipulations did not consistently reproduce the extension in temporal binding metrics seen in behavioural studies of ASD. While we did observe a decrease in binding metrics following a decrease in NMDA and vice versa, observations from clinical studies have frequently reported broadened temporal integration, across various paradigms (Foss-Feig et al., 2010; Stevenson, Siemann, et al., 2014). This discrepancy likely reflects our simplified implementation of NMDA dynamics, which does not incorporate developmental changes in receptor subunit composition (Monyer et al., 1994), NMDA-reliant inhibitory pathways (Carlén et al., 2012), or interactions with synaptic scaffolding proteins implicated in ASD (Coley & Gao, 2019; Duffney et al., 2013). Models incorporating these additional biological mechanisms may be better suited to

capture the relationship between NMDA dysfunction and alterations in integration metrics, in neurodevelopmental disorders.

Experimental predictions and translational implications. The three perturbations explored here shed light on the distinct computational signatures that lead to disruptions in multisensory integration in autism. The findings from this study separate three different phenotypes: a uniform extension in both spatial and temporal windows, which resembles how neurons retain activity over time in Fragile-X models (He et al., 2017); a similar trend following disinhibition, with expanded spatiotemporal windows accompanying degraded precision metrics, resembling PV-interneuron suppression experiments which show prolonged responses (Seybold et al., 2012); and a bidirectional trend reflecting NMDA conductance, which explains why some individuals show narrow temporal windows while others show broadened windows (prolonged GluN2B currents; Martínez-Lázaro et al., 2025). This framework aligns with cross-model surveys that show distinct connectivity clusters. Brain-wide mapping across 16 ASD mutations segregates into hypo- and hyper-connectivity groups, while cortical recordings identify elevated E/I ratios in *Fmr1*, *Cntnap2*, and 16p11.2 deletion models against reduced ratios in *MeCP2*-null mice (Antoine et al., 2019; Banerjee et al., 2016; Zerbi et al., 2021). These mechanistic subtypes argue against treating ASD as monolithic (Just et al., 2012; Lawson et al., 2014; Rubenstein & Merzenich, 2003), instead supporting stratification by binding-window profiles and circuit biomarkers for targeted interventions.

The various perturbation phenotypes explored in this study help to connect divergent findings relating to multisensory processing. The temporal binding window expansion under reduced adaptation and diminished feedforward inhibition aligns with behavioural studies showing autistic children perceive the sound-induced flash illusion at ~600 ms asynchronies, roughly double the ~300 ms neurotypical range (Foss-Feig et al., 2010). This temporal expansion also correlates with speech-in-noise difficulties, where autistic listeners struggle when consonants and vowels are temporally separated by background noise dips (Tsuji & Imaizumi, 2024). Our spatial findings offer a mechanistic account for conflicting peripersonal space reports: while most studies document expanded interpersonal distance preferences in autism (Farkas et al., 2023), Cascio et al. (2008) found smaller, sharper tactile boundaries in adults with ASD. Our model suggests that such diminution could arise from NMDA disturbances

leading to a relative enhancement in centre-surround contrast. This NMDA phenotype also matches the "tunnel vision" enhancement described by Robertson et al. (2013), where spatial attention shows steeper eccentricity fall-off despite broad temporal windows. On the other hand, reduced adaptation widens the spatial window in the model, which, if this generalized to peripersonal space coding, would bias comfortable (peripersonal) distances outward; together with the opposite spatial narrowing seen under NMDA decreases, such contrasting shifts could help account for the individual differences in interpersonal distances reported in ASD (Givon-Benjio et al., 2024). At cellular levels, adaptation-driven expansion mirrors Fragile-X data showing abolished spike-frequency adaptation from reduced K⁺ currents (Luque et al., 2024), while disinhibition manipulations parallel impaired multisensory gain in Shank3 mice with weakened GABAergic inhibition (Pagano et al., 2023). The remaining discrepancy, the decrease in temporal windows contrasting with predominantly expanding temporal windows in humans, suggests additional factors like developmental timing or top-down modulation selectively gate temporal binding in vivo. Nevertheless, the model's responses that do align with biological findings could be investigated further to argue for targeted therapeutic avenues.

Each perturbation mechanism generates specific differences in the responses that could be experimentally validated. Kv7 channels have been highlighted as critical mediators of neuronal adaptations in biological systems; after-hyperpolarisation currents mediated by Kv7 channels regulate excitability and produce early spike-frequency adaptation (Luque et al., 2024). The channel opener retigabine has been shown to restore adaptation via a reduction in spiking frequency and thus, acute or chronic retigabine may reverse the uniform spatial and temporal window enlargement we observed following adaptation reduction (Stas et al., 2016). With regard to feedforward perturbations, deletion of the SHANK3 gene in PV interneurons causes circuit hyper-excitability and abnormal behaviours that can be rescued by enhancing GABAergic signalling (Pagano et al., 2023). Our findings suggest that opto- or chemo-activation of PV cells in Shank3 mice will contract the model matched >300ms temporal binding window towards control values. A similar prediction has already been tested in Fragile-X mice, where boosting interneuron activity after the critical period brought sensory thresholds back to baseline (Kourdougli et al., 2023). In line with such inhibition associated deficits, in vivo micro-iontophoresis of GABA in rat SC has been shown to reduce visual responses and tighten the inhibitory surround (Murthy &

Humphrey, 1999; Razak & Pallas, 2007). Our simulations similarly predict a similar trend: increasing GABAergic center surround strength will narrow spatial binding. Lastly, recent findings have shown that the GlyT1 inhibitor PF-03463275 enhances neuroplasticity by augmenting NMDAR co-agonism in humans (D'Souza et al., 2018; Surti et al., 2023). The current findings therefore indicate that GlyT1 blockade (or other glycine-site agonists) in NMDA-hypofunction mouse lines will broaden the temporal-binding window into the optimal ~200 ms range.

Model scope and architectural refinements. While our spiking model captures core principles of collicular multisensory processing, it is important to consider the simplifications made in this work that limit its validity. We simplified fine layering of superficial SC and excluded cortico-tectal feedback loops, even though visual and prefrontal projections to these strata have been shown to gate cross-modal responses and decision speed in behaving animals (Cuppini et al., 2010; Dash et al., 2018). These top-down signals reshape both spatial and temporal binding windows (Felch et al., 2016). For instance, anterior cingulate suppression lengthens localisation latencies while broadening temporal integration windows, potentially modulating the three phenotypes we modelled (Vázquez, Maulhardt, et al., 2024; Vázquez, Peña-Flores, et al., 2024). Beyond architectural constraints, each perturbation (adaptation reduction, feed-forward disinhibition, NMDA scaling) was tested in isolation, whereas experimental work shows that adaptation and inhibition interact, and multiple synaptic defects frequently co-occur in ASD models and clinical populations (Jiang et al., 2022; Monday et al., 2023; Natan et al., 2017). Moving past the network, the stimulus environment itself employed isolated Gaussian signals rather than the continuous, competing sensory streams that bombard the SC in natural scenes, where stimulus reliability and congruence fluctuate moment-to-moment (Choi et al., 2018). Additionally, we omitted specific neuromodulatory systems, such as endogenous serotonin which scales SC gain and can flip multisensory enhancement into suppression (Tang & Trussell, 2017), and cholinergic input from the pedunculopontine nucleus which sharpens spatial tuning and accelerates saccades (Wolf et al., 2015).

These abstractions, however, represent deliberate methodological choices that enhance interpretability while establishing a validated foundation for systematic perturbation. By focusing on intermediate/deep SC layers, we isolated the canonical, bottom-up rule set (spatial coincidence, inverse-effectiveness) intrinsic to deep SC

circuitry (Stein & Stanford, 2008), making perturbation effects unambiguous; top-down influences can later be incorporated as a separate attentional control layer, following computational frameworks advocated in recent SC modelling reviews (Bisley & Mirpour, 2019; Veale et al., 2017). The present model therefore shows that local SC synaptic dynamics are sufficient within a deliberately bottom-up architecture to generate the reported binding phenotypes. But this does not imply that these mechanisms are uniquely responsible in biology, where cortico-tectal feedback and task context are also likely to shape multisensory integration. We simplified the stimulus paradigm to parallel psychophysics studies that use minimal stimuli like the sound-induced flash illusion to reveal key temporal-binding rules before adding naturalistic complexity (Shams et al., 2000). Accordingly, we interpret comparisons to human TBW/SBW data at the qualitative rather than fitting absolute values, because published estimates vary with task design, stimulus class and extra-collicular influences not represented in the present SC model. Our parameter sweeps are therefore intended to identify plausible mechanistic regimes instead of precise biological parameter values. Once validated, competitive computations observed under dynamic scenes can be layered onto this base architecture. The isolation of individual perturbations follows factorial design principles in computational biology (Murphy & Miller, 2003), establishing the single factor baselines necessary to quantify synergies, particularly relevant given that adaptation and inhibition interact multiplicatively in the cortex. This further suggests that our single factor results represent lower bound shifts that would amplify when combined, mirroring stronger clinical phenotypes (Girirajan et al., 2010). Similarly, excluding neuromodulatory inputs keeps the parameter space tractable while preserving the excitatory-inhibitory core; these signals act largely as multiplicative gates that can be added as a state dimension to capture learning effects without re-tuning the entire synaptic layout (Ferguson & Cardin, 2020; Martinez-Trujillo & Treue, 2004). Collectively, these choices create an interpretable and falsifiable framework while mapping a clear trajectory for incorporating cortico-tectal loops, compound perturbations, naturalistic stimuli, and state-dependent modulation in future iterations (Basso & May, 2017).

Conclusion. Here we modelled a spiking neural instantiation of the SC to investigate how different synaptic and cellular changes produce the multisensory defects seen in ASD. A detailed computational investigation of the network revealed that reduced

neuronal adaptation, diminished feedforward inhibition and altered NMDA conductances each leave distinct marks on spatiotemporal binding, yet can converge on similar behavioural outcomes like enlarged temporal windows. This diversity in mechanisms might explain why individuals across the autism spectrum share certain sensory difficulties while differing in other metrics. The model generates solid predictions that can be tested pharmacologically and ontogenetically, which offers a path from circuit level perturbations to targeted interventions.

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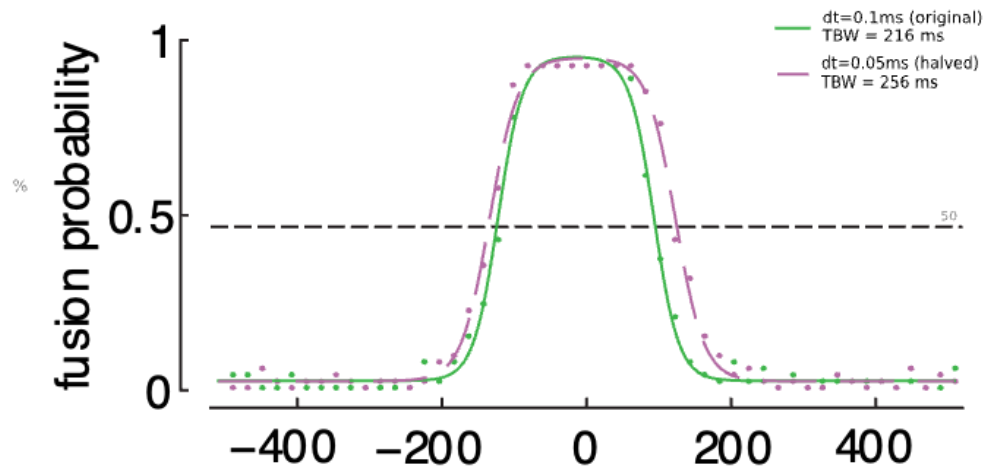
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Supplementary Figures



Supplementary Figure 1. Sensitivity analysis of temporal binding. Fusion probability as a function of audio-visual onset asynchrony for control. Vertical green solid lines indicate TBW fit when time step is 0.1 ms and magenta dashed lines indicate fit when time step is halved to 0.05 ms; negative asynchronies indicate visual-leading stimuli. Error bars indicate $\pm\text{SEM}$ across ten network instantiations.